

LAVERTON, Australia

WMO: 944490

Lat: 28.613S

Lon: 122.424E

Elev: 1525

StdP: 13.90

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	36.9	39.3	12.0	10.7	72.9	15.9	13.0	75.3	24.0	61.3	21.6	60.7	4.9	80	0.548

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.3	105.4	65.3	102.4	64.9	99.3	64.4	74.4	84.9	72.1	84.8	70.3	84.5	12.3	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.8	124.3	79.1	69.0	112.8	76.7	66.3	102.7	75.9	39.1	83.9	36.9	84.4	35.2	83.9	99.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.4	21.5	20.0	DB	31.5	110.2	1.8	2.2	30.1	111.7	29.1	113.0	28.0	114.2	26.7	115.8	(4)
(5)			WB	29.3	77.7	2.3	6.5	27.7	82.3	26.3	86.1	25.0	89.8	23.3	94.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	68.8	82.9	80.8	75.5	70.1	61.2	54.8	53.9	57.0	63.5	71.0	75.7	80.0		
	DBStd	12.19	7.86	7.89	7.53	6.16	6.22	5.18	5.28	6.33	6.92	7.82	7.50	7.39		
	HDD50	42	0	0	0	0	1	12	20	8	1	0	0	0		
	HDD65	1249	2	1	8	21	153	308	345	261	111	32	6	1		
	CDD50	6904	1019	863	791	602	348	156	142	224	405	652	771	931		
	CDD65	2634	556	443	334	172	35	2	1	13	66	218	326	468		
	CDH74	33532	7816	5791	3823	1579	301	15	24	191	886	2658	4129	6319		
	CDH80	18045	4754	3363	1903	598	57	0	1	32	303	1254	2156	3624		
(14) Wind	WSAvg	11.1	13.1	12.9	12.1	10.1	9.2	9.0	8.8	9.8	10.9	12.1	12.6	13.0		
(15) Precipitation	PrecAvg	9.7	0.8	1.6	1.1	0.9	1.0	1.0	0.8	0.6	0.4	0.4	0.6	0.7		
	PrecMax	20.7	7.0	9.2	7.1	3.4	3.2	4.6	2.6	2.2	1.7	2.2	3.5	2.7		
	PrecMin	4.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
	PrecStd	4.0	1.4	2.0	1.4	0.9	0.8	0.9	0.8	0.6	0.4	0.5	0.6	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	110.2	108.4	101.6	94.9	85.2	76.0	77.1	84.1	91.7	99.4	104.5	106.9		
		MCWB	67.8	65.0	63.8	63.5	58.9	55.3	55.7	56.6	57.7	60.8	62.9	65.2		
	2%	DB	105.6	104.5	97.9	90.2	80.9	71.7	72.9	79.5	87.2	95.4	99.2	103.3		
		MCWB	66.4	65.2	63.9	61.8	57.9	53.7	52.9	55.3	57.3	60.5	62.4	63.9		
	5%	DB	102.4	100.8	94.4	86.6	77.2	68.7	69.6	75.3	83.5	91.5	95.4	99.6		
		MCWB	66.1	65.4	63.3	60.9	57.0	52.7	51.7	53.9	56.6	60.1	61.9	63.6		
	10%	DB	98.5	96.2	90.4	83.0	73.6	65.7	66.1	71.1	79.2	87.5	91.6	95.7		
		MCWB	65.8	65.7	63.5	60.3	56.0	51.8	50.7	52.9	55.6	59.4	61.0	63.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	79.1	77.0	73.2	70.0	65.3	60.0	60.3	68.9	67.9	71.5	70.9	72.5		
		MCDB	89.1	86.1	84.1	77.9	71.7	66.3	69.6	71.1	74.5	86.5	82.5	86.6		
	2%	WB	73.9	74.6	70.5	67.1	61.8	57.7	56.6	58.6	60.5	64.1	67.4	70.2		
		MCDB	85.3	85.1	80.7	77.9	73.6	65.6	65.8	69.1	77.8	81.7	83.2	85.2		
	5%	WB	71.6	72.6	68.6	65.4	60.0	55.9	54.5	55.9	58.3	62.0	65.4	68.4		
		MCDB	87.6	83.9	81.5	76.8	70.7	63.6	63.8	70.2	78.2	83.2	83.9	85.2		
	10%	WB	69.7	70.3	66.8	63.8	58.3	54.0	52.6	53.9	56.6	60.5	63.9	66.3		
		MCDB	88.4	83.3	82.2	75.0	69.1	61.6	63.0	68.5	75.9	82.8	84.5	87.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	23.3	21.7	20.4	20.7	20.4	19.9	22.1	23.3	24.3	24.0	23.7	23.6		
		MCDBR	26.9	26.4	25.6	25.6	23.8	22.9	26.2	27.9	29.9	28.8	27.6	27.1		
		MCWBR	7.8	7.6	8.2	9.1	9.9	10.9	12.8	13.7	12.5	10.5	9.0	8.5		
		MCDBR	22.9	20.3	19.6	19.9	18.6	17.7	21.0	23.6	25.7	25.0	24.6	23.7		
	5% WB	MCWBR	9.4	8.6	8.7	9.3	9.0	9.8	11.8	13.9	12.0	10.3	9.4	8.9		
(40) Clear-Sky Solar Irradiance	taub		0.343	0.345	0.326	0.312	0.289	0.277	0.276	0.283	0.303	0.323	0.331	0.331		
	taud		2.568	2.583	2.608	2.623	2.642	2.659	2.642	2.604	2.538	2.501	2.520	2.562		
	Ebn at Noon		317	311	307	297	290	287	292	302	309	314	318	321		
	Edh at Noon		34	33	30	27	24	22	23	27	32	35	36	35		
(44) All-Sky Solar Radiation	RadAvg		2446	2189	1901	1546	1255	1067	1177	1520	1942	2233	2461	2574		
	RadStd		216	179	134	87	79	67	80	82	81	134	107	139		

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(46)
(47)	Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page